upgrading your own infrastructure is easier than getting your franchisees to upgrade theirs, for instance. ISPs that have their own infrastructure are also able to guarantee better services, since they are in full control of quality. However, taking the plunge to network an area is an expensive proposition, and they need to think more than twice before laying out more cabling.

We aren't going to get more penetration this way, indicates Naresh Ajwani, VP Projects at Sify Broadband: "There are 120 million cable TV connections in India just waiting to be exploited." We'll get deeper to the penetration dilemma later, but there's some food for thought in that...

Finally, the people giving you your connection may be franchisees of the ISPs, though many ISPs prefer to establish their own branch offices and give you service themselves.

The Purse-strings

So *why* is bandwidth so expensive? There are a number of factors to consider here, and many of them depend on each other to a point where it's like listening to the chicken-and-egg argument.

The first, and most obvious, is the cost of the infrastructure itself. Connecting a country like India is no small matter—says A S Oberai, Director, IOL Broadband, "India is actually quite exceptional in its degree of connectivity. VSNL has around 12,000 km of optical fibre running across India—and that's *just* VSNL. A lot of countries aren't that well off. I'm told that 18 lakh *villages* are connected by fibre-optic cable—that's one of the greatest things achieved in India."

All commendable and everything, but we still didn't have a clear answer—but everyone we spoke to was quick to point out that prices *are* falling. Today, you get a 256 Kbps unlimited connection for the same price that you got a 64 Kbps connection a few years ago; if you're on an MTNL or BSNL connection, you're paying the same price for your 2 Mbps connection that you did for your 256 Kbps pre-January! And yet...

They key is the return on investment (ROI)—with penetration the way it is in the country, ISPs need to make enough money from subscriptions to offset their investments. Bandwidth *is* an expensive commodity—You Telecom CEO E. V. S Chakravarthy calls it perishable: "We pay for bandwidth by the day, so if any of it goes unused, we make a loss." If you're on a limited data transfer plan, you should be able to relate to this instantly—this is why you can't carry your remaining megabytes forward to the next month.

Penetration is crucial, then—when more people start adopting broadband, the economies of scale will come into play, and the cost per connection will fall. It isn't the only thing, however—while we fight the monster of penetration, there's something else that will drive prices down further...

"It's not that bandwidth is much cheaper abroad—but the demand is such that it's easier for ISPs to make their money back."

A. S. Oberai
Director, IOL Broadband

The Local Bandwidth Funda

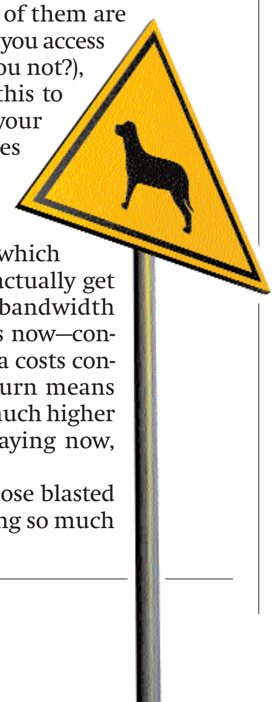
Consider your home or office LAN—the costs involved in maintaining it aren't particularly high. There's the one-time cost of cabling, servers, and routers, and the recurring costs of repairing the odd broken cable, the power to keep the servers running, and so on. While these aren't costs to be scoffed at, they aren't daunting in the big picture—to you, the bandwidth you use when accessing another computer on the network is free. Costs only come into play when you're accessing the Internet, and companies pay a lot of money for high-speed (usually those measured in Mbps) unlimited connections.

Take it a step higher, and the story is similar. Try to picture India as one massive LAN. Accessing sites (servers) that are on this network is cheap—you're using local bandwidth. But then, how many of the sites you visit are hosted in India? "Our dependence on international bandwidth is too much," Ajwani points out, "and prices will drop only when more datacentres are established in India."

All the data that comes to you has travelled from the US or Europe through an NSP's broadband pipe—a pipe that costs them a lot of money to own, thus costing ISPs a lot of money to buy bandwidth off; so you pay more for accessing, say, Yahoo! than someone in the US does.

And that's saying a lot—Yahoo!, Orkut and Google India are the top three sites visited by Indians—and none of them are hosted in India. Every time you access these sites (and how can you not?), it costs money. Compare this to the ideal situation where your most frequently-used sites have their own mirrors within the country. You'll be using the international pipes less often, which means that your ISP can actually get away with buying less bandwidth form the NSP than it does now—connecting to a server in India costs considerably less—which in turn means that you can actually get much higher speeds for what you're paying now, perhaps even cheaper!

Now you know why those blasted friends in the US are getting so much for so little!



Broadband Technology

Wireless broadband connections are still to make their presence felt; today, you get to choose between Internet over cable or ADSL (Asymmetric Digital Subscriber Line). You'll find the latter, most notably, with MTNL, BSNL and Airtel.

Theoretically, cable wins the speed war—it's capable of up to 30 Mbps, though this depends on how well-maintained your connection is. In contrast, ADSL connections can reach a theoretical maximum of 10 Mbps. Then again, the speed you get over your cable connection depends on shared bandwidth—if there are too many users in your area, and your cable provider hasn't allocated enough bandwidth, you'll get a poor connection. The DSL connection, however, is all yours—your bandwidth depends only on the quality of the line.

If you've got a telephone line coming to your house, you're already ADSL-ready—you'll get your connection over that telephone line. With cable, you have the inconvenience of having your house wired.

Finally, with cable connections, upload and download speeds are the same, so you'll see caps on total data transfer. ADSL, being *asymmetric*, can give you different speeds for both. For example, you can download at up to 2 Mbps on an MTNL or BSNL connection, but your upload speed is capped at 256 Kbps—reflecting your normal use. Moreover, there's only a download cap!